

GABRIEL BAZALAR

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EDUCATION

Purdue University

Bachelor of Science in **Electrical Engineering (BSEE)**

West Lafayette, IN | **Expected May 2026**

School Ambassador, Semester Honors (Fall 2024)

Relevant Coursework: Power Electronics, Electromechanical Motion Devices, Signal Analysis, Feedback-Control Design, Microprocessors Interfacing, ASIC Design, Advanced C Programming, Python for Data Science.

INDUSTRY EXPERIENCE

Research Assistant, Wireless Sensing Circuits and Systems Lab

West Lafayette, IN | **January 2025 – Present**

- Developing a **non-invasive Battery Management System (BMS)** for eVTOLs, integrating electromagnetic and current sensors with microcontrollers and an **ECU** to estimate State of Charge (SoC) and State of Health (SoH).
- Implementing **Digital Signal Filtering** techniques to reduce white Gaussian noise, enhancing **ADC/DAC** conversion accuracy and ensuring robust **CAN and I2C** communication.

Summer Research Internship, Flex Laboratory SURF

West Lafayette, IN | **May 2024 – December 2024**

- Designed a low-voltage **piezoelectric energy harvesting PCB** using the LTC3588 IC, integrating over 100 ceramic and polymer sensors embedded in concrete structures.
- Performed impedance matching and analysis using an AIM UHF **impedance analyzer** and oscilloscope, ensuring efficient power transfer and consistent analog readings for measurements of concrete structural strength (5 kHz to 1 MHz).

Hardware Engineering Intern, Calite Inmobiliaria, Gerencia y Construcción

Lima, Perú | **April 2023 – August 2023**

- Tested **IoT controllers** to ensure optimal performance in security cameras, intercoms, and elevator communication systems; deployed Wi-Fi networks using Ubiquiti UniFi to enhance system reliability.

PROJECTS

CAN Communication Board

- Implemented **CAN** communication with the STM32L432KCU6 for the **BMS** team of the Purdue Electric Racing SAE Team, prioritizing design for manufacturability (DFM).
- Tested the emergency Chargin Shutdown Circuit for sending and receiving critical battery information.

Self-Driving Interfacing System

- Developed a system controller using STM32 NUC for SEVCON gen4 motor controller, outputting **PID-controlled** acceleration and pneumatic brakes.
- Utilized 48V synchronous rectification **DC-DC** converter for powering 5-12V computers and electronics.

Video Game Console

- Emulated the video game Piano Tiles on an **STM32** microprocessor, interfacing EEPROM, SD card reader, audio amplifier, analog joystick, and 7-segment displays for full console experience.
- Utilized the **CMSIS** framework in Platform.IO for direct modification of timer interrupts and microprocessor registers.

Lunar Lander Computer System

- Designed a Lunar Lander simulator for **ICE40H8K FPGA** to calculate altitude, velocity, thrust, and landing success, performing Static Timing Analysis (STA) using Siemens **Questa Sim**.

LEADERSHIP

Latin Autonomous Projects Club Founder (LAPs)

May 2024 – Present

- Founded** an engineering club to support Latino students in developing intelligent systems and robotic projects, fostering innovation in Embedded Hardware and Machine Learning.

ECE Student Ambassador (ECEA)

January 2024 – Present

- Planned, managed, and represented** the College of Electrical Engineering at events, allocating a semester budget exceeding \$8,000 to fund student recruitment, innovative projects, and student-led initiatives.

Teaching Assistant (UTA)

January 2024 – August 2024

- Mentored** over 100 students in circuit simulation and PCB prototyping, improving capstone project outcomes through hands-on guidance with **LTSpice, KiCad, and Keysight oscilloscopes**

SKILLS AND INTERESTS

Programming: C, Python, RISC-V Assembly, System Verilog, MATLAB Simulink/QUARC

CAD Tools: Altium Designer, LTSpice, KiCad, Questa Sim, Synopsys VCS, 3DsMax, Fusion 360

Certifications: IBM – ML with TensorFlow, NVIDIA – AI on Jetson Nano, IPA'D – Advanced 3DsMax

Languages: Spanish (native), English (fluent)